IPL Death-Overs Analyst

Understanding guide: the whole project in simple language

1. The problem, in plain words

In a T20 cricket match, the last five overs (overs 16 to 20) are called the death overs. Batters attack hard, so runs flow fast, but a bowler who takes wickets or bowls dot balls here can still win the match. This project asks one sharp question: which bowlers actually perform best in the death overs? It answers with numbers, not reputation, using every ball bowled in the IPL data we downloaded.

2. The dataset

Source: Cricsheet, https://cricsheet.org/downloads/ipl_csv2.zip - public data, no login needed. One row is one ball bowled. After combining all matches we had 295,732 deliveries from 1,243 matches, with season labels from 2007/08 to 2026. Of those, 67,173 deliveries were death-over deliveries in regulation innings. Each row tells us the match, season, innings, ball number, batting and bowling team, striker, bowler, runs off the bat, wides, no-balls, byes, legbyes, and the type of dismissal if a wicket fell.

3. The workflow, step by step

Step 1: fetch_data.py downloads the Cricsheet zip and stacks all per-match CSV files into data/deliveries.csv. Step 2: analysis.py cleans the data (see section 5) and keeps only death overs. Step 3: it groups the death overs by bowler and computes matches, legal balls, runs conceded, wickets, dot balls, economy and dot-ball percentage. Step 4: it keeps only bowlers with at least 100 legal death balls, so tiny samples cannot top the lists. Step 5: it draws three charts and saves everything to a SQLite database, so the same rankings can be checked in SQL with queries.sql.

4. Definitions you must know

Death overs: overs 16 to 20 of innings 1 or 2 only. Super overs (innings 3 and above) are excluded. The raw ball column reads 0.1 for the first ball of over 1, so over number = int(ball) + 1. Legal ball: no wide and no no-ball. Runs conceded by the bowler: runs off the bat + wides + no-balls. Byes and legbyes are NOT charged to the bowler, because the bowler did not cause them. Bowler wicket: any dismissal except a run out, because a run out is the fielders' work. Dot ball: a legal ball with no run off the bat and no bye or legbye. Economy: runs conceded divided by legal balls, multiplied by 6, i.e. runs per over.

5. Every main function: what it does, why, and what it outputs

fetch_data.py - download_zip(url, dest): downloads the zip, or reuses it if it is already on disk. Why: re-running the project should not re-download 7 MB every time. Output: the zip file path. fetch_data.py - combine_matches(raw_dir, output): reads every per-match CSV except the *_info.csv metadata files and stacks them into one CSV. Why: 1,243 small files are impossible to analyse one by one. Output: data/deliveries.csv plus a printed count of matches, rows and seasons. analysis.py - clean_data(df): adds over_number, is_death, is_legal, bowler_runs, is_bowler_wicket and is_dot. Why: every later calculation depends on these six columns being defined exactly once, so Python and SQL can never disagree. Output: a cleaned DataFrame. analysis.py - bowler_death_stats(death): groups death overs by bowler and computes matches, deliveries, legal balls, runs, wickets, dots, economy and dot-ball percentage, then flags bowlers with 100+ legal balls as qualified. Why: one groupby replaces dozens of manual loops. Output: a ranked

DataFrame (main() then writes it to data/death_bowler_stats.csv). analysis.py - save_charts(qualified): draws the economy top 15, the wickets vs economy scatter, and the dot-ball top 15. Why: a chart shows the trade-off between wickets and economy faster than a table. Output: three PNG files in charts/. analysis.py - save_database(df, stats): writes the cleaned deliveries and the bowler stats into data/ipl.db. Why: SQL practice on the same data, and proof the Python numbers match the SQL numbers. Output: the SQLite database file. The two main() functions are the entry points: fetch_data.py main() runs download_zip, extracts the zip and runs combine_matches; analysis.py main() reads deliveries.csv, runs clean_data and bowler_death_stats, writes data/death_bowler_stats.csv, prints the top-10 tables, and runs save_charts and save_database.

6. Key findings (from the real data)

156 bowlers bowled at least 100 legal death balls. Best economy: Sohail Tanvir at 6.82 runs per over, but on only 103 balls in 10 matches, so treat that with caution. The high-volume economy leader is SP Narine at 7.25 over 1,105 balls in 142 matches. Most death wickets: B Kumar (Bhuvneshwar Kumar, listed as B Kumar in the dataset) with 119, then DJ Bravo with 115, then SL Malinga with 108 at an economy of only 7.86. Best dot-ball percentage: Noor Ahmad at 40.4 percent (151 balls), then Narine at 37.4 percent over 1,105 balls. The honest answer to the project question: nobody dominates both lists. Malinga and Narine are the two bowlers who are elite on wickets AND economy, which is why they are the names to defend in an interview.

7. How to explain this project in an interview (60 seconds)

Say this: 'I wanted to find which bowlers actually perform in the IPL death overs, so I downloaded ball-by-ball data from Cricsheet - about 296 thousand deliveries from 1,243 matches. I defined death overs as 16 to 20, counted only legal balls, charged the bowler only for runs off the bat plus wides and no-balls, and required 100 legal death balls before ranking anyone. Then I ranked bowlers by economy, wickets and dot-ball percentage in both Python and SQL, and the two matched. The finding was that wicket-takers usually leak runs, and only Malinga and Narine were elite on both lists.' Then stop and let them ask questions.

8. Five likely interview questions and answers

Q1. Why did you choose overs 16 to 20 as death overs? A: That is the standard last-five-overs phase in a 20-over match, when batters attack hardest and bowling under pressure matters most. Q2. Why the 100-ball minimum? A: Without it, a bowler with one good over would top the economy list. Sohail Tanvir leads at 6.82 but on only 103 balls, while Narine holds 7.25 over 1,105 balls, which is far more trustworthy. Q3. Why do you exclude byes and legbyes from the bowler's runs? A: Those runs are not the bowler's fault - the ball beat the batter and the keeper. Charging them to the bowler would punish good balls. Q4. Why do you exclude run outs from bowler wickets? A: A run out is created by the fielders, not by the delivery, so counting it would credit the bowler for someone else's work. Q5. Why do the top wicket-takers have worse economies? A: Attacking bowling takes wickets but also concedes boundaries, so there is a trade-off, which the wickets-vs-economy chart shows clearly. Malinga and Narine are special because they beat that trade-off.